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4. (a) Mention different functions of materials management & its advantage. 2

(b) Explain purchase model with instantaneous replenishment without shortages. 7

(c) Write functions of stores & different systems instores. 7

(d) Describe ABC analysis & its procedure. 7

5. (a) Mention different stages of inspection. 2

(b) State objectives of Quality Control in Industry & mention benefits of Statistical Quality Control. 7

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(c) Explain the significance of O.C. Curve with figure. 7

(d) Discuss basic principles of JIT and also discuss Push/Pull production system. 7

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B. E. (Eighth Semester) Examination, April-May 2016

(Old Scheme)

(Mechanical, Mechatronics Branch)

PRODUCTION MANAGEMENT

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all questions. Part (a) is compulsory in each question. Attempt any two parts from (b), (c) and (d).

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1. (a) Classify Production System. 2

(b) Discuss briefly the steps involved in new product development. 7

(c) Discuss about make or buy decision & write different approaches for Make or Buy Decision. 7

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(d) Explain the concept of Break Even Analysis with proper formulae & graph. 7

2. (a) What is Sales Forecasting? 2

(b) Name variou forecasting techniques & state the factors based on which forecasting technique is selected. 7

(c) Explain Linear Regression method of forecasting. 7

(d) The details of sales turnover of a fertilizer company for the period 2008-2014 are given in the following table. Compute the estimated sales for the year 2015. 7

Year	2008	2009	2010	2011	2012	2013	2014
Sales (in crore)	39	48	65	78	95	91	112

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3. (a) Name the phases in Production Planning & Control. 2

(b) Explain the concept of MRP & terminologies involved in doing MRP calculations. 7

(c) Write in detail objectives & functions of material handling. 7

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(d) Requirement of an item has been projected in the beginning inventory of 30 units. The carrying cost per unit per week is Rs. 2·50. The cost per set-up if Rs. 250. The lead time to assemble the item is one week. 7

Projected Requirements

Period	0	1	2	3	4	5	6	7	8	9	10	11	12
Projected Requirements	–	30	50	40	–	15	20	20	–	10	–	15	50

(i) Develop an EOQ solution and calculate total relevant cost.

(ii) Develop an MCP solution and calculate total relevant cost.

(iii) Develop a POQ solution and calculate total relevant cost.

(iv) Develop an LUC solution and calculate total relevant cost.

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(v) Develop a PPB solution and calculate total relevant cost.